

SENTINEL AI — DEEPPFAKE INVESTIGATION REPORT

INVESTIGATION ID: INV-AUTH-1790318048 — DIGITAL EVIDENCE EXAMINATION

1. INVESTIGATION METADATA

Investigation ID:	INV-AUTH-1790318048	Case Status:	CASE_UNDER_INVESTIGATION
Case Title:	Multi-Role Evidence Authorization & Processing Audit	Evidence Items:	3
Investigator / Owner:	Standard Case Owner	Assigned Expert:	Investigator Alpha
Incident Date:	N/A	Analysis Date:	2026-09-25 06:34 UTC
Created Date:	2026-09-25 12:04 UTC	Model Version:	Sentinel AI V1.7-A Dual-Head

2. EXECUTIVE SUMMARY

Total Evidence Analyzed	Tampered Classifications	Authentic Classifications	Localization Maps
3	3	0	3 / 3

3. DETAILED EVIDENCE EXAMINATION

EVIDENCE ITEM E-001: 348cc618734d407c96bf4e5f0247be26.jpg

Evidence ID:	50	Uploaded By:	Standard Case Owner (Case Owner)
MIME Type:	image/jpeg	File Size:	41.20 KB
Upload Timestamp:	2026-09-25 06:34 UTC	Analysis Timestamp:	2026-09-25 06:34 UTC
SHA-256 Hash:	be85d1ac3b4dc5572f3983c4c36b78ecf67437a51ffbe30dda5affa954f0d980		



Figure 1: Original Evidence Image



Figure 2: Sentinel AI Heatmap Overlay

Metric	Sentinel AI V1.7-A Value	Threshold Criteria
Classification Result	TAMPERED / MANIPULATED MEDIA	Threshold: 0.40
Tampered Probability	75.0%	Range: [0.0% - 100.0%]
Authentic Probability	25.0%	Range: [0.0% - 100.0%]
Spatial Localization	Heatmap Artifact Generated	Threshold: 0.35

EVIDENCE ITEM E-002: 02c15da21b1f47899a7daf42227ea52e.jpg

Evidence ID:	51	Uploaded By:	Investigator Alpha (Investigator)
MIME Type:	image/jpeg	File Size:	41.20 KB
Upload Timestamp:	2026-09-25 06:34 UTC	Analysis Timestamp:	2026-09-25 06:34 UTC
SHA-256 Hash:	be85d1ac3b4dc5572f3983c4c36b78ecf67437a51ffbe30dda5affa954f0d980		



Figure 1: Original Evidence Image



Figure 2: Sentinel AI Heatmap Overlay

Metric	Sentinel AI V1.7-A Value	Threshold Criteria
Classification Result	TAMPERED / MANIPULATED MEDIA	Threshold: 0.40
Tampered Probability	75.0%	Range: [0.0% - 100.0%]
Authentic Probability	25.0%	Range: [0.0% - 100.0%]
Spatial Localization	Heatmap Artifact Generated	Threshold: 0.35

EVIDENCE ITEM E-003: f69c9eafa9994d009abca3e625e96d44.jpg

Evidence ID:	52	Uploaded By:	Standard Case Owner (Case Owner)
MIME Type:	image/jpeg	File Size:	41.20 KB
Upload Timestamp:	2026-09-25 06:34 UTC	Analysis Timestamp:	2026-09-25 06:34 UTC
SHA-256 Hash:	be85d1ac3b4dc5572f3983c4c36b78ecf67437a51ffbe30dda5affa954f0d980		



Figure 1: Original Evidence Image



Figure 2: Sentinel AI Heatmap Overlay

Metric	Sentinel AI V1.7-A Value	Threshold Criteria
Classification Result	TAMPERED / MANIPULATED MEDIA	Threshold: 0.40
Tampered Probability	75.0%	Range: [0.0% - 100.0%]
Authentic Probability	25.0%	Range: [0.0% - 100.0%]
Spatial Localization	Heatmap Artifact Generated	Threshold: 0.35

4. MODEL METHODOLOGY & TECHNICAL DETAILS

Architecture Overview: Sentinel AI V1.7-A is a frozen dual-head deep learning architecture optimized for image deepfake detection and pixel-level spatial localization.

Preprocessing Stream: Input images are normalized and passed through a dual-stream RGB backbone (EfficientNet-B0) paired with a high-pass Laplacian residual filter bank capturing high-frequency forensic noise signatures.

Dual Heads:

- *Classification Head:* Computes softmax probability across authentic vs. tampered classes using a calibrated decision threshold of **0.40**.
- *Localization Head:* Generates pixel-level feature heatmaps highlighting manipulated regions using a spatial activation threshold of **0.35**.

5. FORENSIC INTERPRETATION & DISCLAIMER

OFFICIAL FORENSIC DISCLAIMER

This report presents automated deepfake classification and spatial localization results produced by Sentinel AI V1.7-A. These findings provide probabilistic indicators based on forensic noise, boundary inconsistencies, and facial feature distributions. This document is intended to assist law enforcement, legal experts, and digital forensic investigators as corroborative evidence. Final evidentiary conclusions should incorporate qualified human expert examination.